

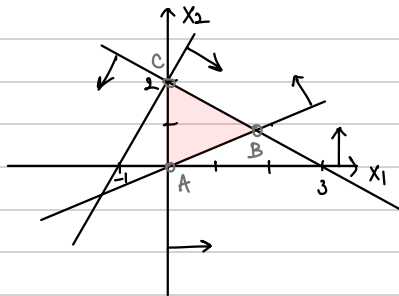
IIE6120 Linear Programming
2025-2 Homework 03
Team B (이정민, 장서현)

[3.2] Consider the following constraints:

$$\begin{aligned} 2x_1 + 3x_2 &\leq 6 \\ -2x_1 + x_2 &\leq 2 \\ x_1 - 2x_2 &\leq 0 \\ x_1, x_2 &\geq 0. \end{aligned}$$

- Draw the feasible region.
- Identify the extreme points, and at each extreme point identify all possible basic and nonbasic variables.
- Suppose that a move is made from the extreme point $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$ to the extreme point $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ in the (x_1, x_2) space. Specify the possible entering and leaving variables.

a.



b.

$$\begin{aligned} 2x_1 + 3x_2 + s_1 &= 6 \\ -2x_1 + x_2 + s_2 &= 2 \\ x_1 - 2x_2 + s_3 &= 0 \\ x_1, x_2, s_3 &\geq 0 \end{aligned} \quad \rightarrow \quad 3 \text{ BVs, } 2 \text{ NBVs.}$$

EP	BVs	NBVs
Point A $(x_1, x_2) = (0, 0)$	$(s_1, s_2, s_3) = (6, 2, 0)$ $(x_2, s_1, s_2) = (0, 6, 2)$ $(x_1, s_1, s_2) = (0, 6, 2)$	$(x_1, x_2) = (0, 0)$ $(x_1, s_3) = (0, 0)$ $(x_2, s_3) = (0, 0)$
Point B $(x_1, x_2) = (12/7, 6/7)$	$(x_1, x_2, s_2) = (12/7, 6/7, 32/7)$	$(s_1, s_3) = (0, 0)$
Point C $(x_1, x_2) = (0, 2)$	$(x_1, x_2, s_3) = (0, 2, 4)$ $(x_2, s_1, s_3) = (2, 0, 4)$ $(x_2, s_2, s_3) = (2, 0, 4)$	$(s_1, s_2) = (0, 0)$ $(x_1, s_2) = (0, 0)$ $(x_1, s_1) = (0, 0)$

c.

$$(x_1, x_2, s_1, s_2, s_3) = (0, 2, 0, 0, 4)$$

possible entering vars: s_1 or s_2

possible leaving vars: x_2 or s_3

[3.5] Solve the following linear programming problem by the simplex method.

At each iteration, identify B and B^{-1} .

$$\begin{array}{llll} \text{Maximize} & 3x_1 & + & 2x_2 & + & x_3 \\ \text{subject to} & 3x_1 & - & 3x_2 & + & 2x_3 & \leq & 3 \\ & -x_1 & + & 2x_2 & + & x_3 & \leq & 6 \\ & x_1, & & x_2, & & x_3 & \geq & 0. \end{array}$$

$$\text{Max } z = 3x_1 + 2x_2 + x_3$$

$$\text{s.t. } 3x_1 - 3x_2 + 2x_3 + x_4 = 3$$

$$-x_1 + 2x_2 + x_3 + x_5 = 6$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

$$c = (3, 2, 1, 0, 0)^T$$

$$A = [a_1, a_2, a_3, a_4, a_5] = \begin{pmatrix} 3 & -3 & 2 & 1 & 0 \\ -1 & 2 & 1 & 0 & 1 \end{pmatrix} \quad b = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$$

① initial basis $B = [a_4, a_5]$

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad B^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$c_B = (0, 0) \quad c_B B^{-1} = (0, 0)$$

$$\bar{b} = B^{-1}b = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

$$x_1 = 0, x_2 = 0, x_3 = 0, x_4 = 3, x_5 = 6$$

$$z_1 - c_1 = c_B B^{-1} a_1 - c_1 = -3 \rightarrow x_1 \text{ enters}$$

$$z_2 - c_2 = c_B B^{-1} a_2 - c_2 = -2 \quad y_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

$$z_3 - c_3 = c_B B^{-1} a_3 - c_3 = -1 \quad = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix} - \begin{bmatrix} 3 \\ -1 \end{bmatrix} x_1 \rightarrow x_4 \text{ leaves}$$

$$x_1 = 1, x_2 = 0, x_3 = 0, x_4 = 0, x_5 = 7$$

② $B = [a_1, a_5]$

$$B = \begin{bmatrix} 3 & 0 \\ -1 & 1 \end{bmatrix} \quad B^{-1} = \begin{bmatrix} 1/3 & 0 \\ 1/3 & 1 \end{bmatrix}$$

$$c_B = (3, 0) \quad c_B B^{-1} = (1, 0)$$

$$\bar{b} = B^{-1}b = \begin{bmatrix} 1/3 & 0 \\ 1/3 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$

$$x_1 = 1, x_2 = 0, x_3 = 0, x_4 = 0, x_5 = 7$$

$$z_2 - c_2 = c_B B^{-1} a_2 - c_2 = -5 \rightarrow x_2 \text{ enters}$$

$$z_3 - c_3 = c_B B^{-1} a_3 - c_3 = 1 \quad y_2 = \begin{bmatrix} 1/3 & 0 \\ 1/3 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$

$$z_4 - c_4 = c_B B^{-1} a_4 - c_4 = 1$$

$$\begin{bmatrix} x_1 \\ x_5 \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} x_2 \rightarrow x_5 \text{ leaves}$$

$$x_1 = 8, x_2 = 7, x_3 = 0, x_4 = 0, x_5 = 0$$

③ $B = [a_1, a_2]$

$$B = \begin{bmatrix} 3 & -3 \\ -1 & 2 \end{bmatrix} \quad B^{-1} = \begin{bmatrix} 2/3 & 1 \\ 1/3 & 1 \end{bmatrix}$$

$$c_B = (3, 2) \quad c_B B^{-1} = (8/3, 5)$$

$$\bar{b} = B^{-1}b = \begin{bmatrix} 2/3 & 1 \\ 1/3 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \end{bmatrix} = \begin{bmatrix} 8 \\ 7 \end{bmatrix}$$

$$x_1 = 8, x_2 = 7, x_3 = 0, x_4 = 0, x_5 = 0$$

$$z_3 - c_3 = c_B B^{-1} a_3 - c_3 = 28/3 > 0$$

$$z_4 - c_4 = c_B B^{-1} a_4 - c_4 = 8/3 > 0$$

$$z_5 - c_5 = c_B B^{-1} a_5 - c_5 = 5 > 0$$

optimal! $\rightarrow z = 38, x_1 = 8, x_2 = 7, x_3 = x_4 = x_5 = 0$

[3.8] Consider a maximization linear programming problem with extreme points x_1, x_2, x_3 , and x_4 , and extreme directions d_1, d_2 , and d_3 , and with an objective function gradient c such that $cx_1 = 4, cx_2 = 6, cx_3 = 6, cx_4 = 3, cd_1 = 0, cd_2 = 0$, and $cd_3 = -2$. Characterize the set of alternative optimal solutions to this problem.

x_2, x_3 can achieve the highest obj. value.
 d_1, d_2 does not change the obj. value.

$$\{x^*: \lambda x_2 + (1-\lambda)x_3\}$$

[3.13] Answer the following questions along with a concise explanation with respect to the linear program to maximize cx subject to $x \in X = \{x: Ax = b, x \geq 0\}$, where A is $m \times n$ of rank $m < n$.

- In a simplex tableau, if $z_j - c_j = -7$ for a nonbasic variable x_j , what is the change in objective value when x_j enters the basis given that the minimum ratio is 3 in the pivot?
- If an extreme point is optimal, then is it possible that not all $z_j - c_j \geq 0$ for an associated basis?
- If there exists a d such that $Ad = 0$, $d \geq 0$, and $cd > 0$, then is the optimal objective value unbounded?
- Let $\bar{x}$ be a feasible solution with exactly m positive components. Is $\bar{x}$ necessarily an extreme point of X ?
- If a nonbasic variable x_k has $z_k - c_k = 0$ at optimality, then can one claim that alternative optimal solutions exist?

a. 21

$$\Delta Z = -(z_j - c_j) \times (\text{minimum ratio}) \\ = -(-7) \times 3 = 21$$

b. False

An extreme point is optimal if and only if $z_j - c_j \geq 0$ for all nonbasic variables j .

c. True.

Let x be a feasible solution.
Then $x + \lambda d$, $\lambda \geq 0$ also becomes a feasible solution
The obj. value becomes $c(x + \lambda d) = cx + \lambda cd$
Since $cd > 0$, the obj. value can increase unboundedly by increasing λ .

d. False

only if the m positive components are linearly independent

e. True

If $z_k - c_k = 0$, this means x_k can be introduced into the basis without changing the obj value. This indicates the existence of alternative optimal solutions.

f. True

x_1 and x_2 being adjacent means that they are made through a single pivot operation. Thus, the corresponding bases differ by one column, making them adjacent bases.

g. True

If there are alternative optimal solutions, there are multiple optimal extreme points. Any convex combination of them is also an optimal solution. $\rightarrow$ There can be more than m positive variables

- If x_1 and x_2 are adjacent points and if B_1 and B_2 are respective associated bases, then these bases are also adjacent. True or false? Explain.
- Is it possible for an optimal solution to have more than m positive variables?
- Suppose that $n = m + 1$. What is the least upper bound on the number of extreme points and feasible bases?
- A p -dimensional polyhedron can have at most p extreme directions. True or false? Explain.
- Let $\bar{x}$ be an extreme point having $(m - 1)$ positive components. Then there are $(p + 1)$ bases associated with this extreme point, where $p = n - m$. True or false? (Assume that $Ax = b$ does not imply any variable to be a constant.) Explain.

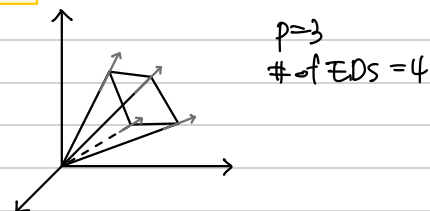
h. $m+1, m+1$

$$\binom{n}{m} = \binom{m+1}{m} = \frac{(m+1)!}{m!(m+1-m)!} = m+1$$

$\downarrow$
feasible bases.

Each feasible basis corresponds to an extreme point. Thus, the maximum of E.P.s $= m+1$.

i. False



j. False

$\bar{x}$ is a degenerate point. We can choose from $n-m+1$ columns to form the basis. $p+1$
Yet, there is no guarantee that $p+1$ are linearly independent. Thus, there can be less than $p+1$ bases.

[3.19] Suppose that we have a basic feasible solution that is nondegenerate. Furthermore, suppose that an improving nonbasic variable enters the basis.

- ① Show that if the minimum ratio criterion for exiting from the basis occurs at a unique index, then the resulting basic feasible solution is nondegenerate. Is this necessarily true if the first solution is degenerate? ②

①

Let x_k be the entering variable and let x_r be the leaving variable

$$\theta = \min_i \left\{ \frac{\bar{b}_i}{y_{ik}} \mid y_{ik} > 0 \right\} = \frac{\bar{b}_r}{y_{rk}}$$

- Basic Variable $x_k > 0$

Proof :

since the initial solution is non-degenerate, $\bar{b}_r > 0$.

and since we only consider $y_{rk} > 0$, $\theta > 0$.

Thus, $x_k > 0$.

- Remaining Basic Variables x_{bi} ($i \neq r$)

Proof :

By θ , $\bar{b}_i$ will be updated $\rightarrow \bar{b}_i' = \bar{b}_i - y_{ik} \cdot \theta$

if $y_{ik} \leq 0$, $\bar{b}_i' = \bar{b}_i - y_{ik} \cdot \theta \geq \bar{b}_i > 0$.

if $y_{ik} > 0$, since θ is the minimum value among $\frac{\bar{b}_i}{y_{ik}}$, $\frac{\bar{b}_i}{y_{ik}} > \theta$.

Thus, $\bar{b}_i' = \bar{b}_i - y_{ik} \cdot \theta > 0$

Since all basic variables in the updated basis are strictly positive, the solution is non-degenerate.

② No

If the first solution is degenerate, this means at least one basic variable is 0.

Let basic variable $x_r = 0$, so $\bar{b}_r = 0$.

Say an improving non-basic variable x_k entered the basis and $y_{rk} > 0$.

The minimum ratio is $\theta = \frac{\bar{b}_r}{y_{rk}} = 0$.

Then x_k would enter as 0 and the remaining basic variable would not change.

The new basis contains $x_k = 0$, resulting in degeneracy.

[3.28] Consider the following problem:

$$\begin{array}{llllllll} \text{Maximize} & -3x_1 & + & 2x_2 & - & x_3 & + & x_4 \\ \text{subject to} & 2x_1 & - & 3x_2 & - & x_3 & + & x_4 \leq 0 \\ & -x_1 & + & 2x_2 & + & 2x_3 & - & 3x_4 \leq 1 \\ & -x_1 & + & x_2 & - & 4x_3 & + & x_4 \leq 8 \\ & x_1, & & x_2, & & x_3, & & x_4 \geq 0. \end{array}$$

$$\text{Max } z = -3x_1 + 2x_2 - x_3 + x_4$$

$$\text{s.t. } 2x_1 - 3x_2 - x_3 + x_4 + x_5 = 0$$

$$-x_1 + 2x_2 + 2x_3 - 3x_4 + x_6 = 1$$

$$-x_1 + x_2 - 4x_3 + x_4 + x_7 = 8$$

$$x_1, x_2, x_3, x_4, x_5, x_6, x_7 \geq 0.$$

Use the simplex method to verify that the optimal objective value is unbounded. Make use of the final simplex tableau to construct a feasible solution having an objective of at least 3500. Make use of the final tableau to construct a direction d such that $cd > 0$.

	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS	ratio
z	1	3	-2	1	-1	0	0	0	0	
x_5	0	2	-3	-1	1	1	0	0	0	
x_6	0	-1	2	-3	0	1	1	0	1	$\frac{1}{2} \rightarrow x_6 \text{ leave, } x_2 \text{ enter}$
x_7	0	-1	1	-4	1	0	0	1	8	8

	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS	ratio
z	1	2	0	3	-4	0	1	0	1	
x_5	0	1/2	0	2	-7/2	1	3/2	0	3/2	
x_2	0	-1/2	1	1	-3/2	0	1/2	0	1/2	
x_7	0	-1/2	0	-3	5/2	0	-1/2	1	15/2	3 $\rightarrow x_7 \text{ leave, } x_4 \text{ enter}$

	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS	ratio
z	1	6/5	0	-5	0	0	1/5	8/5	13	
x_5	0	-1/5	0	-5	0	1	4/5	7/5	24	
x_2	0	-4/5	1	-2	0	0	1/5	3/5	5	
x_4	0	-1/5	0	-2	1	0	-1/5	2/5	3	3

x_3 can increase unboundedly $\rightarrow$ UNBOUNDED

• feasible solution

$$\bar{x} = (0.5, 0, 3, 24, 0, 0)^T$$

$$d = (0, 2, 1, 2, 5, 0, 0)^T$$

$$z = 13 + 5x_3 > 3500 \rightarrow \text{Let } x_3 = 700$$

$$x' = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 5 + 700 \cdot 2 \\ 0 + 700 \cdot 1 \\ 3 + 700 \cdot 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1405 \\ 700 \\ 1403 \end{pmatrix} \rightarrow z' = 2 \cdot 1405 - 700 + 1403 = 3513 > 3500$$

• direction

$$\begin{pmatrix} x_5 \\ x_2 \\ x_4 \end{pmatrix} = \begin{pmatrix} 24 \\ 5 \\ 3 \end{pmatrix} - \begin{pmatrix} -5 \\ -2 \\ -2 \end{pmatrix} x_3 \rightarrow d = (d_1, d_2, d_3, d_4, d_5, d_6, d_7)^T = (0, 2, 1, 2, 5, 0, 0)^T$$

$$cd = (-3 \ 2 \ -1 \ 1 \ 0 \ 0 \ 0) \begin{pmatrix} 0 \\ 2 \\ 1 \\ 2 \\ 5 \\ 0 \\ 0 \end{pmatrix} = 5 > 0$$

[3.35] Show that in the simplex method if a variable x_j leaves the basis, it cannot enter the basis in the next iteration.

Let x_k be the entering variable and x_j , more specifically x_{Br} , be the leaving variable.
Let y_{rk} be the pivot of this iteration.

① Min.

It is implied that $z_k - c_k > 0$ and $y_{rk} > 0$.

$$(z_{Br} - c_{Br})_{\text{new}} = (z_{Br} - c_{Br})_{\text{old}} - \frac{1}{y_{rk}} (z_k - c_k) \\ = 0 + \frac{c_k - z_k}{y_{rk}}$$

$$c_k - z_k < 0 \text{ \& } y_{rk} > 0 \rightarrow (z_{Br} - c_{Br})_{\text{new}} < 0$$

$\therefore$ Cannot be selected in next iteration

② Max.

It is implied that $z_k - c_k < 0$ and $y_{rk} > 0$.

$$(z_{Br} - c_{Br})_{\text{new}} = (z_{Br} - c_{Br})_{\text{old}} - \frac{1}{y_{rk}} (z_k - c_k) \\ = 0 + \frac{c_k - z_k}{y_{rk}}$$

$$c_k - z_k > 0 \text{ \& } y_{rk} > 0 \rightarrow (z_{Br} - c_{Br})_{\text{new}} > 0$$

$\therefore$ Cannot be selected in next iteration

[3.37] Consider the linear programming problem: Maximize $\mathbf{c}\mathbf{x}$ subject to $\mathbf{A}\mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$, where $\mathbf{A}$ is an $m \times n$ matrix of rank m . Suppose that an optimal solution with basis $\mathbf{B}$ is at hand. Further suppose that $\mathbf{b}$ is replaced by $\mathbf{b} + \lambda \mathbf{d}$ where λ is a scalar and $\mathbf{d}$ is a fixed nonzero vector of dimension m . Give a condition such that the basis $\mathbf{B}$ will be optimal for all $\lambda \geq 0$.

An optimal solution implies that ① $z_j - c_j \geq 0 \quad \forall j \in J : \text{n.b.v.s.}$
② $x_B \geq 0$

For ①, $z_j - c_j = c_B B^{-1} a_j - c_j \rightarrow$ not influenced by $\mathbf{b}$.
 $\therefore$ the change to $\mathbf{b} \rightarrow \mathbf{b} + \lambda \mathbf{d}$ does not matter

For ②, $x_B = B^{-1} \mathbf{b} \rightarrow x_B' = B^{-1}(\mathbf{b} + \lambda \mathbf{d})$
 $\geq 0 \quad \geq 0$ must hold for the solution to be optimal

$$x_B' = B^{-1}(\mathbf{b} + \lambda \mathbf{d}) \\ = B^{-1} \mathbf{b} + \lambda B^{-1} \mathbf{d}.$$

Here $B^{-1} \mathbf{b} \geq 0$. $\lambda \geq 0$. Thus for $x_B' \geq 0$, the condition $[B^{-1} \mathbf{d} \geq 0]$ must hold

$$\therefore B^{-1} \mathbf{d} \geq 0$$

[3.44] Consider the following linear program:

$$\begin{aligned} \text{Minimize} \quad & -x_1 - 2x_2 + x_3 \\ \text{subject to} \quad & 2x_1 + x_2 + x_3 \leq 6 \\ & 2x_2 - x_3 \leq 3 \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

$$\text{MM } z = -x_1 - 2x_2 + x_3$$

$$\text{s.t. } 2x_1 + x_2 + x_3 + x_4 = 6$$

$$2x_2 - x_3 + x_5 = 3$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0.$$

a.

$$\text{I} \quad z \quad x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad \text{RHS}$$

$$z \quad 1 \quad 1 \quad 2 \quad -1 \quad 0 \quad 0 \quad 0$$

$$x_4 \quad 0 \quad 2 \quad 1 \quad 1 \quad 1 \quad 0 \quad 6 \quad 6$$

$$x_5 \quad 0 \quad 0 \quad 2 \quad -1 \quad 0 \quad 1 \quad 3 \quad 1.5 \rightarrow x_5 \text{ leave, } x_2 \text{ enter}$$

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad N = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & -1 \end{bmatrix} \quad B^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad B^{-1}N = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & -1 \end{bmatrix} \quad C_B B^{-1} = (0, 0) \\ z_j - C_j = (1, 2, -1)$$

$$\text{II} \quad z \quad x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad \text{RHS}$$

$$z \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad -1 \quad -3$$

$$x_4 \quad 0 \quad 2 \quad 0 \quad 3/2 \quad 1 \quad -1/2 \quad 9/2 \rightarrow x_4 \text{ leave, } x_1 \text{ enter}$$

$$x_2 \quad 0 \quad 0 \quad 1 \quad -1/2 \quad 0 \quad 1/2 \quad 3/2$$

$$B = \begin{bmatrix} x_4 & x_2 \\ 1 & 1 \\ 0 & 2 \end{bmatrix} \quad N = \begin{bmatrix} x_1 & x_3 & x_5 \\ 2 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \quad B^{-1} = \begin{bmatrix} 1 & -1/2 \\ 0 & 1/2 \end{bmatrix} \quad B^{-1}N = \begin{bmatrix} 2 & 3/2 & -1/2 \\ 0 & -1/2 & 1/2 \end{bmatrix} \quad C_B B^{-1} = (0, -1) \\ z_j - C_j = (1, 0, -1)$$

$$\text{III} \quad z \quad x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad \text{RHS}$$

$$z \quad 1 \quad 0 \quad 0 \quad -3/4 \quad -1/2 \quad -3/4 \quad -2/4$$

$$x_1 \quad 0 \quad 1 \quad 0 \quad 3/4 \quad 1/2 \quad -1/4 \quad 9/4$$

$$x_2 \quad 0 \quad 0 \quad 1 \quad -1/2 \quad 0 \quad 1/2 \quad 3/2$$

$$B = \begin{bmatrix} x_1 & x_2 \\ 1 & 1 \\ 0 & 2 \end{bmatrix} \quad N = \begin{bmatrix} x_3 & x_4 & x_5 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \quad B^{-1} = \begin{bmatrix} 1/2 & -1/4 \\ 0 & 1/2 \end{bmatrix} \quad B^{-1}N = \begin{bmatrix} 3/4 & 1/2 & -1/4 \\ -1/2 & 0 & 1/2 \end{bmatrix} \quad C_B B^{-1} = (-1/2, -3/4) \\ z_j - C_j = (-3/4, -1/2, -3/4)$$

b.

$$\frac{\partial z}{\partial x_1} = -3/4 : x_1 \text{ must decrease by } 3/4 \text{ to maintain feasibility}$$

$$\frac{\partial z}{\partial x_2} = 0 : x_2 \text{ is not affected by } x_4$$

$$\frac{\partial z}{\partial x_5} = 3/4 : \text{one unit increase in } x_5 \text{ leads to } 3/4 \text{ in } z.$$

$$\frac{\partial z}{\partial x_4} = \begin{pmatrix} -3/4 & -1/2 & 1/4 \\ 1/2 & 0 & -1/2 \end{pmatrix} = A : \text{one unit increase in nbvs leads to changes in bvs according to the elements of } A.$$

c.

$$C'_B = (C'_1, C'_2) = (-1 + \Delta_1, -2 + \Delta_2)$$

$$B^{-1} = \begin{pmatrix} 1/2 & -1/4 \\ 0 & 1/2 \end{pmatrix}$$

$$C_B B^{-1} = \left(-\frac{1}{2} + \frac{\Delta_1}{2}, -\frac{3}{4} - \frac{\Delta_1}{4} + \frac{\Delta_2}{2} \right)$$

$$z_4 - C_4 = -\frac{1}{2} + \frac{\Delta_1}{2} \leq 0$$

$$z_5 - C_5 = -\frac{3}{4} - \frac{\Delta_1}{4} + \frac{\Delta_2}{2} \leq 0$$

$$z_3 - C_3 = (C'_B B^{-1} a_3 - C_3) = \left(-\frac{1}{2} + \frac{\Delta_1}{2} \right) - \left(-\frac{3}{4} - \frac{\Delta_1}{4} + \frac{\Delta_2}{2} \right) - 1 \\ = -\frac{3}{4} + \frac{3\Delta_1}{4} - \frac{\Delta_2}{2} \leq 0$$

$$\therefore \Delta_1 \leq 1, 3\Delta_1 - 2\Delta_2 \leq 3, \Delta_1 - 2\Delta_2 \geq -3$$

- Find an optimal solution by the simplex method. At each iteration identify $B, N, B^{-1}, B^{-1}N, C_B B^{-1}$, and the $(z_j - c_j)$ -values.
- At optimality, find $\partial x_1 / \partial x_3, \partial x_2 / \partial x_4, \partial z / \partial x_5, \partial x_B / \partial x_N$, where x_4 and x_5 are the slack variables. Interpret these values.
- Suppose that c_1 is changed from -1 to $-1 + \Delta_1$, and c_2 is changed from -2 to $-2 + \Delta_2$. Find the region in the (Δ_1, Δ_2) space that will maintain optimality of the solution you obtained in Part (a).
- Suppose that a new activity x_6 is considered. Here $c_6 = -4, a_{16} = 2$, and $a_{26} = 4$. Is it worthwhile to produce the activity? If your answer is yes, find the new optimal solution.
- Suppose that b_1 is changed from 6 to $6 + \Delta$. Find the range of Δ that will maintain optimality of the basis found in Part (a).
- Make use of the final tableau to represent the column a_3 as a linear combination of a_1 and a_2 .

d.

$$z_6 - c_6 = c_B B^{-1} a_6 - c_6 = (-1/2, -3/4) \begin{pmatrix} 1 \\ 4 \end{pmatrix} - (-4) = 0 \rightarrow \text{not worthwhile} \text{ does not change obj. value.}$$

e.

$$B^{-1}b = \begin{bmatrix} 1/2 & -1/4 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} 6+4 \\ 3 \end{bmatrix} = \begin{bmatrix} 9/4 + 4/2 \\ 3/2 \end{bmatrix} \rightarrow 9/4 + 4/2 \geq 0, \quad 4 \geq -4.5$$

f.

$$B^{-1}a_3 = \begin{bmatrix} 3/4 \\ -1/2 \end{bmatrix}, \quad a_3 = [a_1 \ a_2] \begin{bmatrix} 3/4 \\ -1/2 \end{bmatrix} = 3/4 a_1 - 1/2 a_2$$

$$\therefore a_3 = 3/4 a_1 - 1/2 a_2$$

[3.48] The starting and current tableaus of a given problem are shown below. Find the values of the unknowns a through n .

Starting tableau

z	x_1	x_2	x_3	x_4	x_5	RHS
1	a	1	-3	0	0	0
0	b 3	2	2	1	0	6
0	-1	2	-1	0	1	1

Current tableau

z	x_1	x_2	x_3	x_4	x_5	RHS
1	0	-1/3	j	k	1 0	n
0	3 1	2/3	2/3	1/3	0	2 2
0	b 0	8 3	-1/3	1/3	1	3 3

$$a=2, b=3, c=2, d=2, e=-1, \\ f=2, g=1, h=0, i=8/3, j=-13/3, \\ k=-7/3, l=0, m=3, n=-4$$

Starting tableau: $BV = \{x_4, x_5\}$

Current tableau: $BV = \{x_1, x_5\}$: $g=1, h=0, l=0$

$\rightarrow x_4$ leave, x_1 enter $\rightarrow b$ becomes the pivot element

$$[b \ c \ d \ | \ 0 \ || \ 6] \div b = [1 \ 2/3 \ 2/3 \ 1/3 \ 0 \ || \ f]$$

$$c/b = 2/3, \ d/b = 2/3, \ 1/b = 1/3, \ 6/b = f$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ c=2 & d=2 & f=2 \end{array}$$

$$[0 \ i \ -1/3 \ 1/3 \ 1 \ || \ m] = [-1 \ 2 \ e \ 0 \ 1 \ || \ 1] + [1 \ 2/3 \ 2/3 \ 1/3 \ 0 \ || \ 2]$$

$$= [0 \ 8/3 \ e+2/3 \ 1/3 \ 1 \ || \ 3]$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ i=8/3 & m=3 & e=-1 \end{array}$$

$$[0 \ -1/3 \ j \ k \ 0 \ || \ n] = -a \times [1 \ 2/3 \ 2/3 \ 1/3 \ 0 \ || \ 2] + [a \ 1 \ -3 \ 0 \ 0 \ || \ 0]$$

$$= [0 \ 1 - \frac{2}{3}a \ -3 - \frac{2}{3}a \ -1/3a \ 0 \ || \ -2a]$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ j=-13/3 & h=-4 & a=2 \end{array}$$